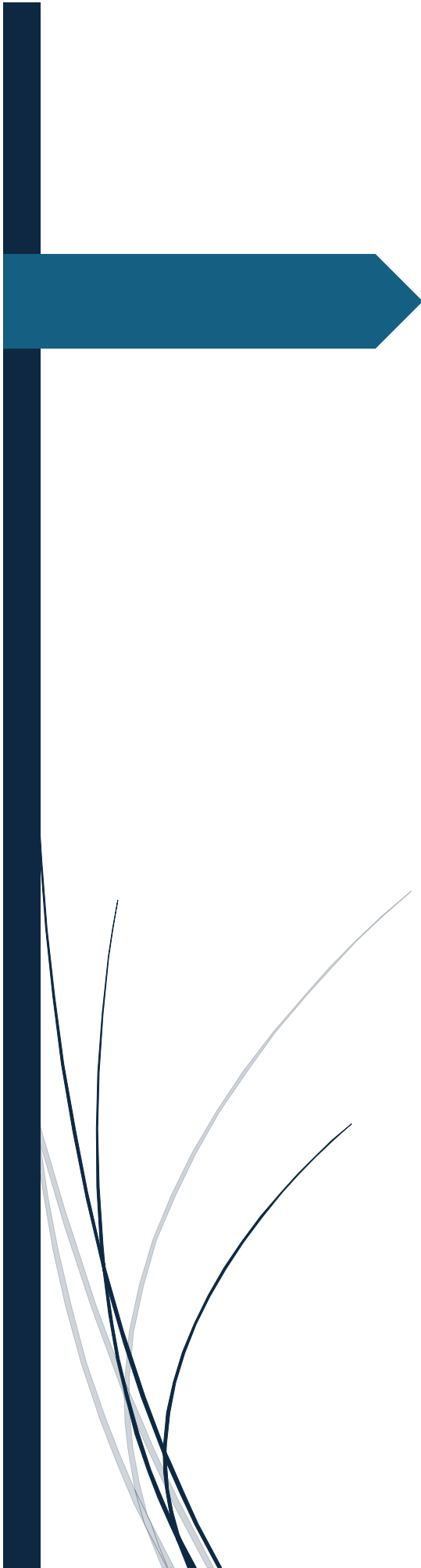
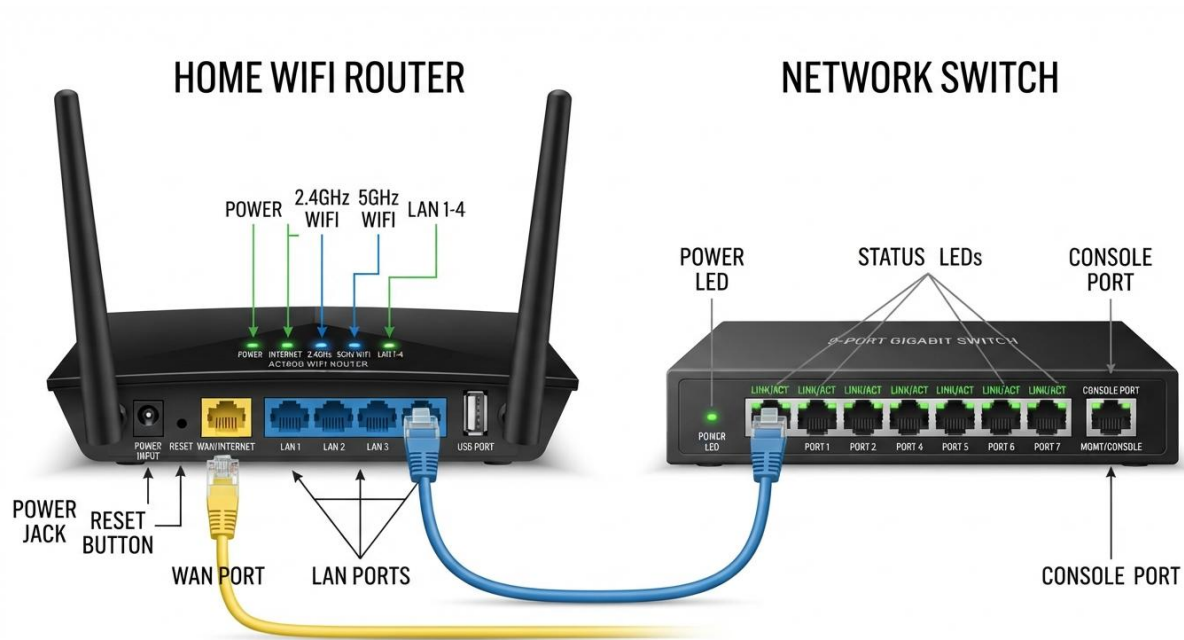




Assignment 2

Routers and switches



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Feature	 Router	 Switch
What it connects	Connects your home to the Internet (outside world).	Connects your devices to each other (inside your home).
How it finds devices	Uses IP addresses (like a street address for your whole house).	Uses MAC addresses (like a unique fingerprint for each device).
Main job	Acts like a security guard to share internet safely.	Acts like a power strip to add more plug-in ports

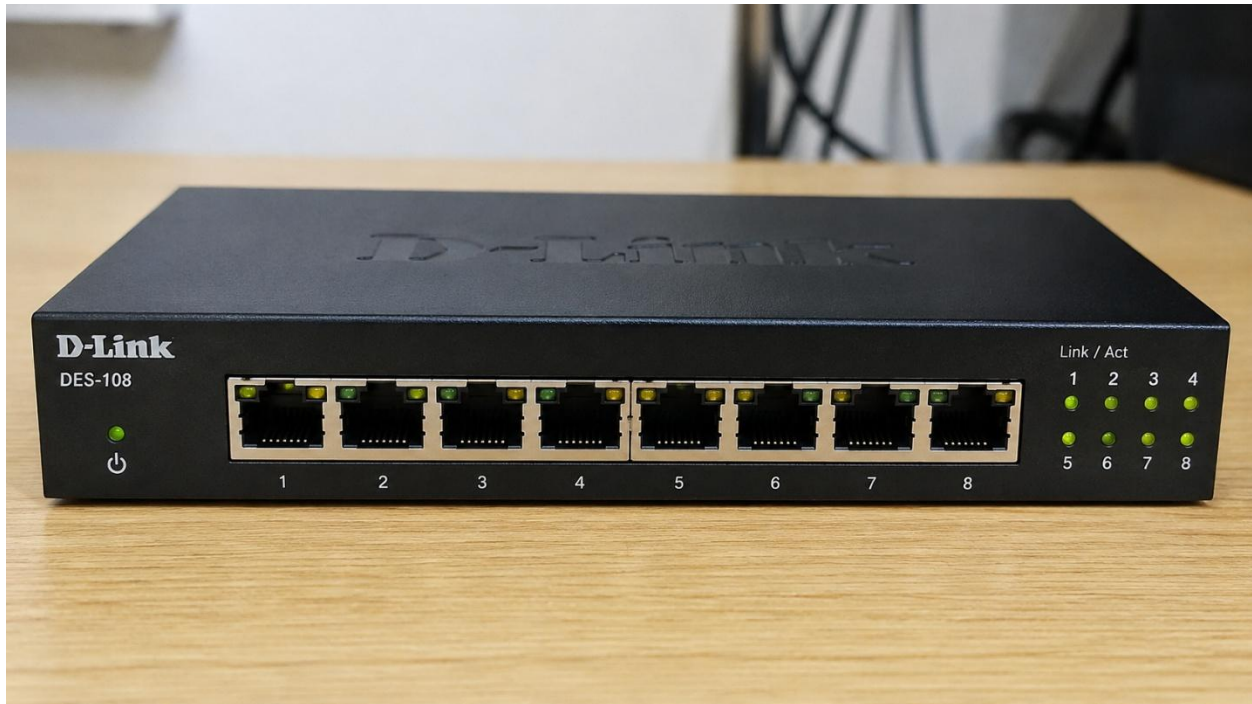
3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

1. Router



- **Power Port:** On the left side. It connects the router to the power adapter
- **Ethernet Ports:** The blue WAN port and four yellow LAN ports are in the middle. They connect the router to the internet and other wired devices.
- **LED Indicators:** On the right side. They show the Power, WAN, LAN, Wi-Fi, and WPS status of the router.

2. Switch



Power Port: Located on the back side of the switch. It connects the switch to the power adapter.

Ethernet Ports: The 8 numbered ports (1–8) are on the front. They connect computers and other network devices using Ethernet cables.

LED Indicators: Located above/near the Ethernet ports. They show whether a port is connected and transmitting data.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

➤ The router connects to the ISP line first because it gives internet access to all PCs and manages the connections. A switch alone can't connect to the internet by itself.

- **Connecting the PCs:**

ISP → Router → Switch → PCs

Since a router usually doesn't have enough ports for PCs, you connect a **switch** to the **router**, and then connect all gaming PCs to that switch using cables.

Why:

- **Router** = gives internet to everyone
- **Switch** = connects all the PCs together and to the router
- **Cables** (not Wi-Fi) = faster and more stable for gaming and streaming